

FRELIGHTSBURG, QC, Canada

WMO: 713730

Lat: 45.0394N

Lon: 72.8561W

Elev: 224

StdP: 98.66

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.2	-21.2	-29.1	0.3	-23.6	-26.1	0.4	-20.6	8.4	-4.1	7.7	-4.7	2.1	270	0.484

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	29.3	21.7	27.8	21.0	26.4	20.1	23.3	27.4	22.3	26.1	21.4	25.0	2.7	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.0	17.1	25.5	21.0	16.1	24.5	20.1	15.2	23.7	70.5	27.6	66.6	26.2	63.4	25.0	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.2	6.2	5.4	-28.4	31.9	3.7	1.3	-31.1	32.8	-33.3	33.6	-35.4	34.3	-38.1	35.2
			-28.6	25.3	3.6	1.2	-31.2	26.1	-33.3	26.8	-35.3	27.5	-37.9	28.4
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Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.1	-8.2	-6.7	-1.6	6.0	13.2	18.0	20.5	19.6	15.7	9.0	2.5	-4.1
	DBStd	11.37	7.72	6.70	6.35	5.14	4.50	3.77	2.92	3.05	4.20	4.93	5.66	6.29
	HDD10.0	2310	564	468	364	141	19	1	0	0	4	81	232	437
	HDD18.3	4327	823	701	619	371	168	52	12	22	99	291	474	694
	CDD10.0	1237	1	0	3	21	119	239	326	297	176	49	7	0
	CDD18.3	214	0	0	0	1	10	41	79	61	21	1	0	0
	CDH23.3	1365	0	0	1	13	83	295	509	347	113	3	0	0
	CDH26.7	264	0	0	0	2	13	67	105	59	18	0	0	0
Wind	WSAvg	2.3	2.6	2.6	2.7	2.7	2.4	2.0	1.9	1.8	2.0	2.3	2.6	2.5
Precipitation	PrecAvg	1147	75	66	76	81	101	110	119	112	107	106	90	102
	PrecMax	1358	172	185	126	166	197	193	167	210	196	184	124	200
	PrecMin	911	32	34	49	18	45	36	78	48	34	43	53	51
	PrecStd	116	30	32	19	38	40	40	26	33	45	44	21	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.5	11.4	19.0	25.4	28.5	31.1	31.3	30.4	29.3	23.6	18.6	11.7
		MCWB	11.4	7.4	13.4	15.9	18.9	23.0	23.9	22.8	22.3	16.8	14.0	9.5
	2%	DB	8.4	7.5	12.8	20.9	25.7	28.7	29.2	28.3	26.2	21.2	15.9	9.0
		MCWB	6.3	4.6	8.4	13.2	17.0	21.0	22.1	21.5	20.5	16.0	12.0	7.1
	5%	DB	4.9	5.0	9.7	17.4	23.3	26.7	27.7	26.8	24.3	18.8	13.5	6.6
		MCWB	3.3	2.6	5.8	10.5	15.9	19.9	21.2	20.6	19.3	14.4	10.3	5.1
	10%	DB	2.2	2.7	7.2	14.4	21.0	24.8	26.2	25.3	22.4	16.5	10.9	4.1
		MCWB	0.6	0.7	4.1	8.8	14.8	18.9	20.4	20.0	18.3	13.0	8.2	2.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.7	8.2	14.2	17.2	20.8	24.2	25.1	24.0	23.0	18.8	15.2	10.1
		MCDB	13.6	10.0	19.6	23.5	25.7	29.1	29.9	28.4	27.3	21.3	17.4	11.4
	2%	WB	6.7	5.0	9.0	14.2	18.9	22.5	23.5	22.7	21.5	16.7	12.7	7.4
		MCDB	7.8	7.1	12.0	19.5	23.0	26.8	27.4	26.5	25.0	20.1	15.2	8.6
	5%	WB	3.2	2.6	6.4	11.6	17.3	21.2	22.4	21.7	20.2	14.9	10.4	5.1
		MCDB	4.5	4.6	8.8	16.1	21.4	25.1	26.0	25.2	23.2	18.0	12.8	6.6
	10%	WB	0.7	0.9	4.3	9.4	15.7	19.9	21.5	20.8	18.9	13.4	8.2	2.6
		MCDB	2.0	2.6	7.1	13.6	19.7	23.2	25.0	24.1	22.0	16.0	10.5	3.9
Mean Daily Temperature Range	5% DB	MDBR	8.8	9.0	8.7	9.6	10.2	9.7	9.4	9.4	9.4	8.2	7.4	7.4
		MCDBR	10.3	10.0	11.8	14.3	12.9	12.1	10.9	10.7	10.8	10.7	10.8	8.8
		MCWBR	9.5	8.2	8.3	8.6	7.1	6.5	5.6	5.6	6.2	6.9	8.2	8.1
	5% WB	MCDBR	10.0	9.7	10.5	12.6	10.9	10.1	9.2	9.3	9.5	9.7	10.0	8.8
		MCWBR	9.4	8.5	8.2	8.5	7.0	6.1	5.4	5.4	6.1	6.9	8.6	8.5
Clear-Sky Solar Irradiance	taub	0.262	0.284	0.319	0.349	0.370	0.418	0.427	0.405	0.365	0.327	0.303	0.281	
	taud	2.381	2.305	2.357	2.392	2.389	2.251	2.234	2.291	2.396	2.510	2.497	2.381	
	Ebn at Noon	865	902	915	912	899	852	839	845	854	848	809	796	
	Edh at Noon	73	96	106	112	116	134	135	123	102	79	67	67	
All-Sky Solar Radiation	RadAvg	1.43	2.33	3.51	4.30	5.05	5.52	5.59	4.99	3.98	2.31	1.42	1.01	
	RadStd	0.17	0.24	0.26	0.52	0.51	0.45	0.42	0.24	0.32	0.23	0.18	0.09	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+62	N/A

Nomenclature: See separate page